Where Small Quantum Classifiers Help, and Where They Don't: Fair Baselines and Real-Hardware Results on IBM Heron

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Abstract

We benchmark small quantum neural networks (QNNs) against classical models under four rules: classical baselines see the same inputs, a strong classical ceiling is always reported, hardware runs are compared with noise-free simulation on identical samples, and anomalies are re-run against the shot-noise expectation. On classical data (two text tasks and three standard datasets), a 4-qubit QNN fed by a sentence-embedding encoder matches but does not clearly beat logistic regression on the same four features (within about one point) (sentiment: 0.758 vs. 0.769; CVE severity: 0.699 vs. 0.692). On IBM's 156-qubit `ibm_marrakesh`, the compiled circuits (depth 37, six two-qubit gates) reproduced noise-free predictions exactly on 4 of 5 datasets; repeated runs attribute the remaining drop to shot noise ($p \approx 0.27$), and a standard fake-backend noise model overstated the bias fivefold. On quantum data, a two-copy QNN separates pure from mixed states with about 32 copies from 2 to 6 qubits, while classical shadows need 128 to 4,096, reproducing the known exponential multi-copy separation at small scale. Code and all result files are public.

1 Introduction

Claims of quantum advantage in machine learning are hard to evaluate when results are reported without a matched classical baseline [1]. We ask a deliberately narrow question: for small, trainable quantum classifiers that run on today's hardware, which tasks show any benefit over classical models, and how much does real-device noise change the answer?

We report two projects. The first trains a 4-qubit QNN on classical data, including two text-classification tasks. The second classifies quantum states directly, where theory predicts separations for learners with quantum memory [2]. Both run end to end on IBM Quantum hardware through the free Open Plan. Our contributions are:

- an evaluation protocol that keeps quantum and classical models on identical inputs and hardware and simulation on identical samples (Section 2);
- evidence that circuit design, not qubit count, dominated accuracy on classical data, and that a well-designed QNN matches but does not clearly beat classical models there (Section 3);
- a repeated-run analysis showing a current Heron-generation device is effectively noise-free for shallow classifiers, and that a commonly used noise model overstates its bias (Section 4);
- a small-scale reproduction of the exponential copy-complexity gap between two-copy QNNs and classical shadows for purity testing (Section 5).

2 Evaluation protocol

1. **Same inputs.** Classical baselines (logistic regression, RBF SVM, and an MLP where noted) receive exactly the features encoded into the circuit.
2. **Classical ceiling.** We also report the best practical classical model, trained on all available data and features.
3. **Same samples on hardware.** Device runs use a test subset to save quantum time; they are compared only with the noise-free simulator on that subset.
4. **Repeat anomalies.** An unexpected hardware result is re-run, and the number of flipped predictions is compared with the expectation from finite sampling.

Rule 3 was not a formality: our first hardware chart placed 40-sample device accuracy beside 200-sample simulator accuracy, which made the device appear better than the simulator on one dataset.

3 Quantum classifiers on classical data

Model. Inputs are reduced to one feature per qubit and scaled to $[0, \pi]$. The circuit repeats, twice, an $R_Y(x_i)$ encoding layer followed by a trainable `RealAmplitudes` block (data re-uploading [3]); the class is the measured value of qubit 0. Weights are trained with L-BFGS-B on cross-entropy, using exact simulated probabilities. We used Qiskit 1.4 [4].

An initial design with the `ZZFeatureMap` encoding [5], all-qubit parity readout and training on sampled counts barely exceeded chance (breast cancer: 0.68). The design above reached 0.92 on the same split, a larger effect than any change in qubit count we tried.

Text tasks. Texts are embedded with all-MiniLM-L6-v2 [6] (384 dimensions) and reduced to four features with partial least squares (PLS), fitted on the full training split. On text, supervised PLS outperformed PCA by about 15 points for a classical model on the same four features. We use Rotten Tomatoes sentence polarity [7] and 12,116 NVD CVE descriptions published January–April 2024, labelled high/critical if the CVSS v3.1 base score is at least 7.0, split by publication date (80/20). The QNN trains on 600 texts.

Task (test size)	Majority	QNN, 4 qubits	LogReg, same 4	LogReg, full embedding
Sentiment (1,066)	0.500	0.758	0.769	0.774
CVE severity (2,424)	0.578	0.699	0.692	0.715

Table 1: Test accuracy on the text tasks. The QNN matches a classical model given the same four features.

Figure 1 extends the comparison to breast cancer, iris (versicolor vs. virginica) and two moons. The QNN is within 0–10 points of the best classical model and never clearly better.

4 Real-hardware behaviour

We ran the trained weights on `ibm_marrakesh` with 1,024 shots per input. After transpilation the circuit has depth 37 and six CZ gates. On identical test subsets (40 inputs; 30 for iris), device

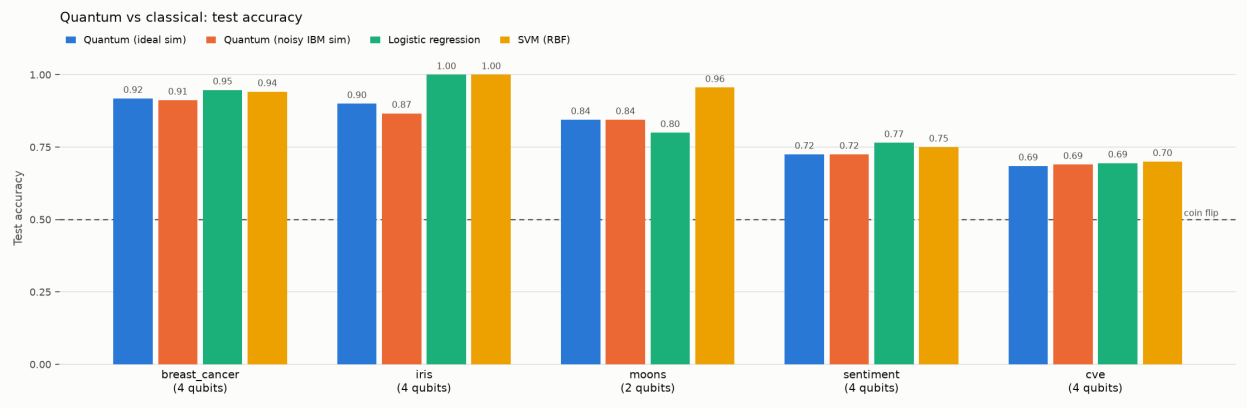


Figure 1: Test accuracy of the QNN (ideal and simulated-noise) and classical models on identical reduced features.

accuracy equalled noise-free accuracy on four of five datasets (Figure 2); CVE severity dropped from 0.725 to 0.650.

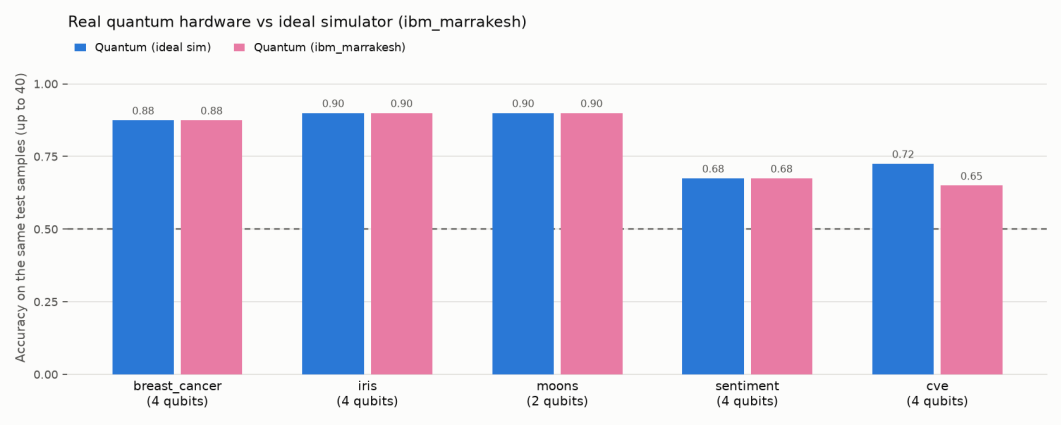


Figure 2: Noise-free simulator vs. `ibm_marrakesh` on the same test inputs.

To test whether the drop reflects device bias, we re-ran the same 40 CVE inputs three times. For an input with noise-free probability p, the chance that N-shot sampling alone crosses the decision boundary is

$$P_{\text{flip}}(p) = \Phi\left(-\frac{|p - \frac{1}{2}|}{\sqrt{p(1-p)/N}}\right). \tag{1}$$

Summed over inputs this predicts 1.15 flips per run. We observed 2, 2 and 1 (accuracies 0.725, 0.675, 0.700): 5 flips against 3.45 expected, a Poisson tail probability of ≈ 0.27. No input flipped in every run, and every flipped input had $|p - \frac{1}{2}| < 0.025$. The mean shift of $P(\text{class } 1)$ was -0.006. The same analysis on Qiskit's `FakeManilaV2` noise model gave a mean shift of -0.030 and two inputs that flipped in every run, overstating the bias of the real device.

5 Quantum classifiers on quantum data

The second project gives models direct access to quantum states. Classical learners receive classical-shadow features [8] from random-Pauli snapshots, or full state vectors where noted.

Entanglement (2–4 qubits). A two-copy QNN separates entangled from product states perfectly in exact simulation; a one-copy QNN stays at chance, as its output is linear in ρ. Classical models on full state vectors fall towards chance as qubits increase (0.69 to 0.54), while the best model on 256 shadow snapshots reaches 0.94–0.99 depending on size.

Phases. For ground states of the transverse-field Ising chain (4–8 qubits) and a $\mathbb{Z}_2$ lattice gauge model, QCNNs [9], generic QNNs and classical models all score 0.96–1.00: no quantum edge.

Purity. Distinguishing Haar-random pure states from random rank-2 mixed states requires estimating a global property. Single-copy strategies need a number of copies growing exponentially with system size, while a two-copy SWAP-type test needs $O(1)$ [2]. Our learned two-copy QNN reaches 90% accuracy with 16–32 copies at every size from 2 to 6 qubits; classical shadows need 128, 256, 512, 1,024 and 4,096 (Figure 3). Training the two-copy model hit a barren plateau [10] at 5 qubits with independent per-pair blocks; sharing one block across copy pairs, or transferring weights trained at 2 qubits, removed it.

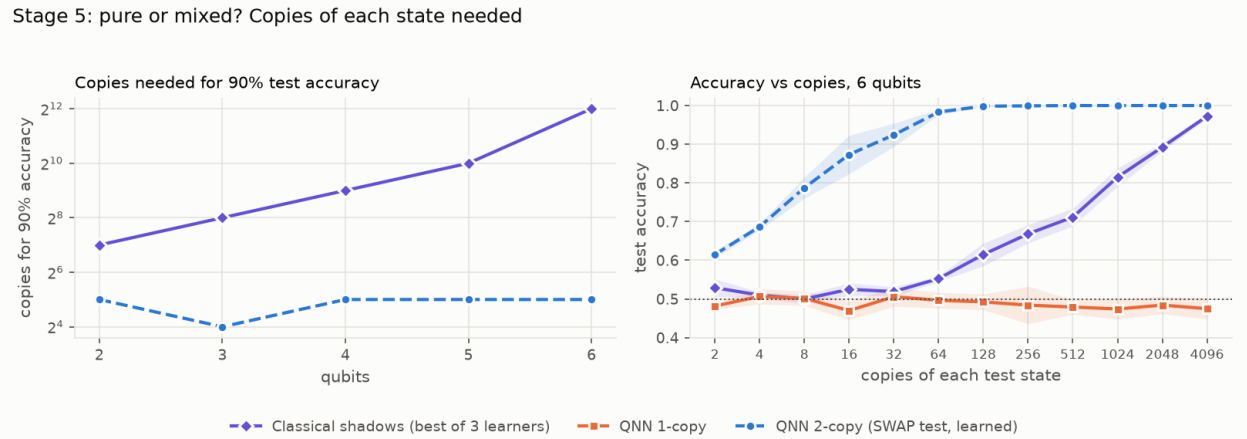


Figure 3: Pure vs. mixed states. Left: copies of each test state needed for 90% accuracy. Right: accuracy vs. copies at 6 qubits (mean and spread over 3 seeds).

6 Discussion and limitations

Across both projects, a quantum edge appeared only where the model could hold two copies of a quantum state at once; this matches theory [2], which we reproduce rather than extend. On classical data the circuit acts as a small classifier on features chosen by a classical encoder, and classical models on the same features do as well. Whether such QNNs can be matched by quantum-inspired classical surrogates [11] on these tasks is left for future work.

Limitations: all simulations are at most 8 qubits and exact, so nothing here is classically hard; device evaluations use 30–40 inputs per dataset, enough for an end-to-end check rather than fine-

grained accuracy claims; CVE severity from descriptions alone approximates CVSS scoring. The two-copy purity test has not yet been run on hardware.

Reproducibility. All code, configurations and result files are in two repositories: `https://github.com/H3XY/AGI` and `https://github.com/H3XY/Entangle`. During this work we found that `qiskit-ibm-runtime` 0.44.1 declares Qiskit 1.x support but fails to transpile for real devices on it; we reported it upstream (Qiskit/qiskit-ibm-runtime#3563).

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